COMPUTER NETWORKS LABORATORY
COMPUTER NETWORK LAB TEST
EXPERIMENTS 1 AND 2

LAB TEST IMPLEMENTATION REPORT

Submitted by

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EXPERIMENT 1

Introduction to Packet Tracer, Peer-to-Peer Communication, Study of Cables and its Color Codes

1.1 Aim

To become familiar with the Cisco Packet Tracer interface, to build a simple peer-to-peer network of two computers connected through a switch, to verify communication between them using the ping command, and to study the types of Ethernet cables and their colour codes.

1.2 Requirements

- Software: Cisco Packet Tracer
- Devices: two PCs (PC-PT) and one Switch-PT
- Cable: copper straight-through Ethernet cable

1.3 Procedure

1. Launch Cisco Packet Tracer and note the workspace, device-selection box and toolbar.
2. Place two PCs (PC0 and PC1) and one switch (Switch0) on the workspace (Figure 1.1).
3. Connect each PC to the switch using a copper straight-through cable. The link lights are orange while the link is coming up (Figure 1.2) and turn green when the connection is up (Figure 1.3).
4. Open PC0, go to Desktop > IP Configuration and assign the IP address and subnet mask given in Table 1.1 (Figure 1.4). Repeat for PC1 (Figure 1.5).
5. Open the Command Prompt on PC0 and run ping 192.168.1.2 to test connectivity with PC1 (Figure 1.6). Repeat from PC1 with ping 192.168.1.1 (Figure 1.7).
6. Observe the packet flow between the PCs in Simulation mode (Figure 1.8).
7. Study the wire colour codes of the straight-through and crossover cables (Tables 1.2 and 1.3).

Table 1.1: IP addressing

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

1.4 Network Setup



Figure 1.1: PC0, Switch0 and PC1 placed on the workspace



Figure 1.2: Links coming up (orange link lights)



Figure 1.3: Devices connected with straight-through cables (all links green)

1.5 IP Configuration

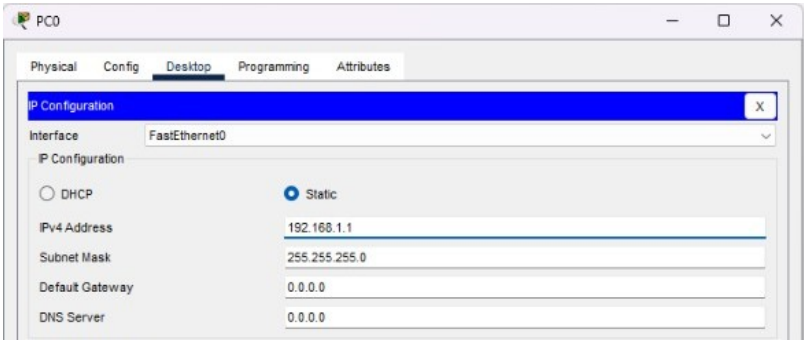


Figure 1.4: IP configuration of PC0 (192.168.1.1)

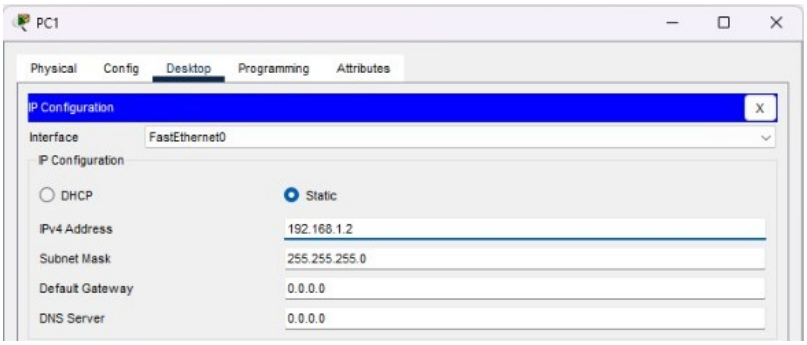


Figure 1.5: IP configuration of PC1 (192.168.1.2)

1.6 Output

The ping command was executed from PC0 to PC1 and from PC1 to PC0. In both cases four packets of 32 bytes were sent and four replies were received.

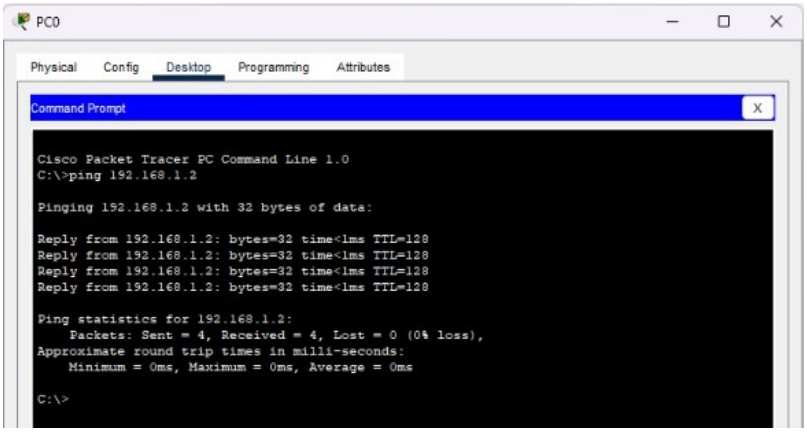


Figure 1.6: Ping from PC0 to PC1 (192.168.1.2)

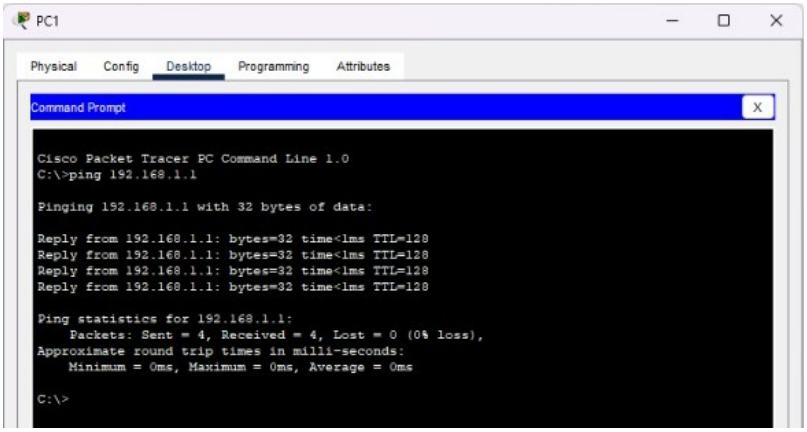


Figure 1.7: Ping from PC1 to PC0 (192.168.1.1)

Parameter	Observed value
Packets sent / received / lost	4 / 4 / 0 (0% loss)
Round-trip time	Minimum 0 ms, Maximum 0 ms, Average 0 ms
TTL	128
Bytes per packet	32

1.7 Simulation Mode



Figure 1.8: Packet transfer from PC0 through Switch0 to PC1 in Simulation mode

1.8 Study of Cables and Colour Codes

Ethernet cables contain four twisted pairs (eight wires) terminated in an RJ-45 connector. The wires are arranged according to the TIA/EIA-568 standard, which has two wiring schemes, T568A and T568B.

Table 1.2: Pin-wise colour codes

Pin	T568A	T568B
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

Table 1.3: Cable types

Cable	Wiring	Used to connect
Straight-through	Same standard (T568B) at both ends	Different devices, e.g. PC to switch, PC to router, switch to router
Crossover	T568A at one end and T568B at the other	Similar devices, e.g. PC to PC, switch to switch, router to router

1.9 Result

The two-PC network was created in Packet Tracer using a switch and straight-through cables. The ping between PC0 and PC1 succeeded in both directions with 0% packet loss, which verifies peer-to-peer communication, and the cable types and colour codes were studied.

EXPERIMENT 2

Implementation of Network Topologies

2.1 Aim

To implement network topologies in Cisco Packet Tracer: a point-to-point connection of two computers using a crossover cable, and a star topology using a switch, and to verify connectivity in each case using the ping command.

2.2 Theory

A network topology is the physical or logical arrangement of the devices in a network. In a bus topology all devices share one backbone cable; in a star topology every device connects to a central switch or hub; in a ring topology each device connects to exactly two neighbours forming a closed loop; and in a mesh topology every device connects to every other device. In this experiment a crossover cable is used to connect two similar devices (PC to PC) directly, and straight-through cables are used to connect each PC to the central switch of the star topology.

2.3 Requirements

- Software: Cisco Packet Tracer
- Devices: two PCs (PC0 and PC1) and one 2960-24TT switch (Switch1)
- Cables: copper crossover cable (PC to PC) and copper straight-through cable (PC to switch)

2.4 Part A: Point-to-Point Connection Using a Crossover Cable

Procedure

1. Place two PCs (PC0 and PC1) on the workspace (Figure 2.1).
2. Select the copper crossover cable and connect the FastEthernet0 port of PC0 to the FastEthernet0 port of PC1 (Figure 2.2).
3. Assign the IP addresses given in Table 2.1 to both PCs (Figures 2.3 and 2.4).
4. Run ping 192.168.10.30 from PC1 and observe the reply (Figure 2.5).

Table 2.1: IP addressing

Device	IP Address	Subnet Mask
PC0	192.168.10.30	255.255.255.0
PC1	192.168.10.50	255.255.255.0



Figure 2.1: PC0 and PC1 placed on the workspace



Figure 2.2: PC0 and PC1 connected with a copper crossover cable

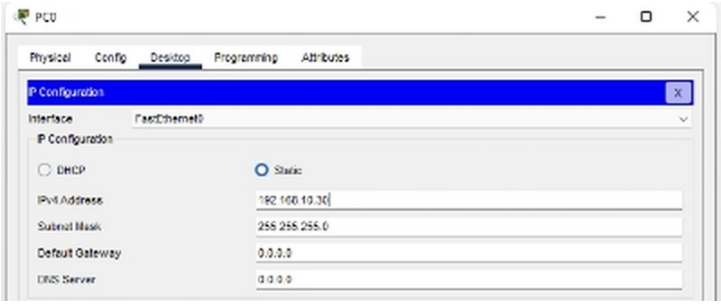


Figure 2.3: IP configuration of PC0 (192.168.10.30)

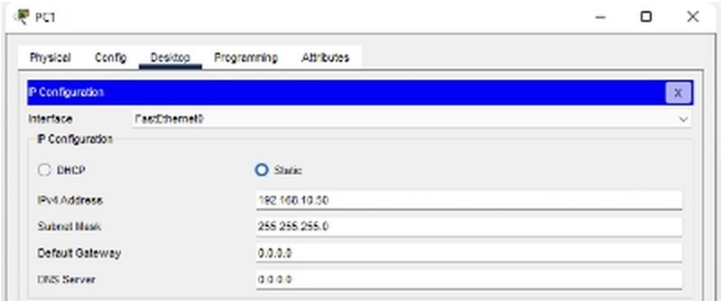


Figure 2.4: IP configuration of PC1 (192.168.10.50)

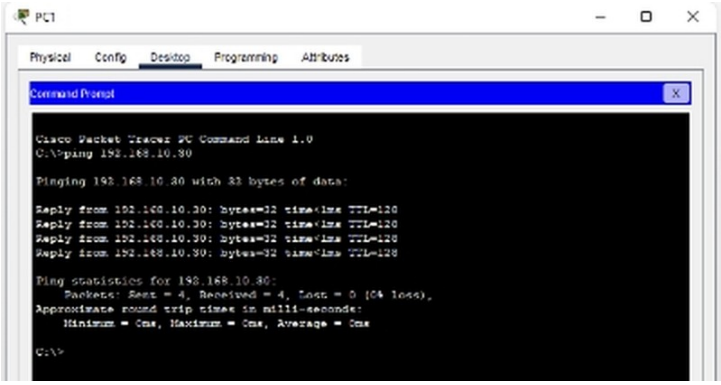


Figure 2.5: Ping from PC1 to PC0 (192.168.10.30) over the crossover cable

Parameter	Observed value
Packets sent / received / lost	4 / 4 / 0 (0% loss)
Round-trip time	Minimum 0 ms, Maximum 0 ms, Average 0 ms
TTL	128
Bytes per packet	32

2.5 Part B: Star Topology Using a Switch

Procedure

1. Add a 2960-24TT switch (Switch1) to the workspace along with PC0 and PC1 (Figure 2.6).
2. Select the copper straight-through cable and connect PC0 to the first free FastEthernet port of the switch, then connect PC1 to the next free port (Figures 2.7 and 2.8).
3. Keep the IP addresses of Table 2.1 on both PCs.
4. Run ping 192.168.10.50 from PC0 and observe the reply (Figure 2.9).



Figure 2.6: Switch1 (2960-24TT), PC0 and PC1 placed on the workspace

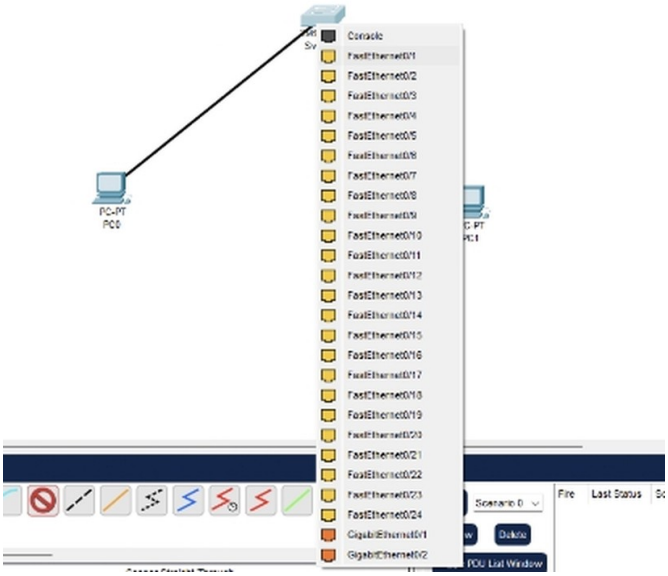


Figure 2.7: Selecting the switch port for PC0 (FastEthernet0/1)

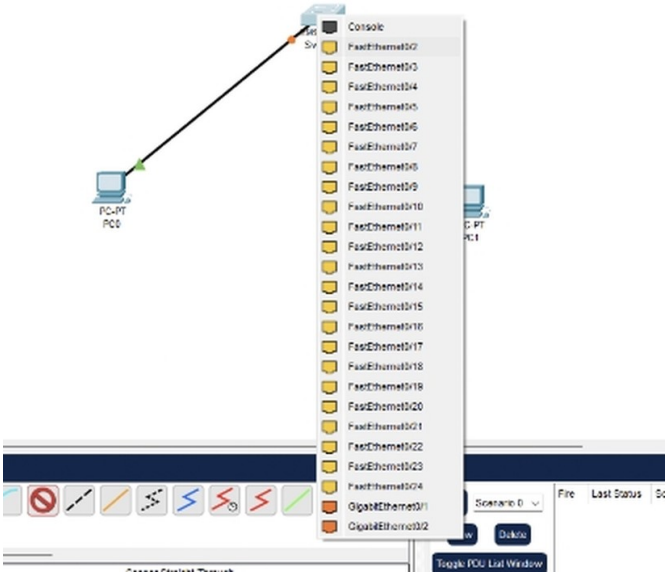


Figure 2.8: PC0 connected; selecting the next free switch port for PC1

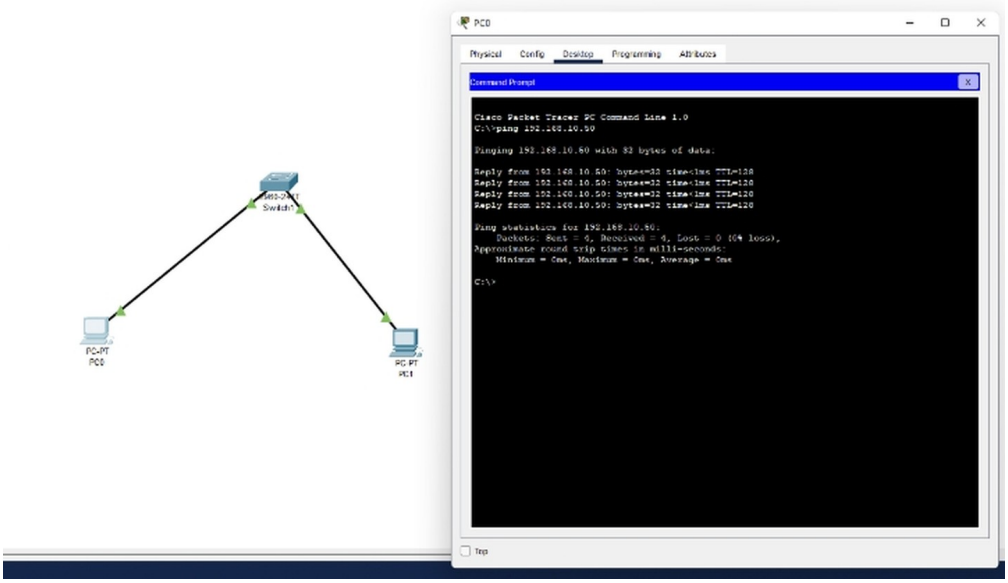


Figure 2.9: Star topology with Switch1 at the centre and ping from PC0 to PC1 (192.168.10.50)

Parameter	Observed value
Packets sent / received / lost	4 / 4 / 0 (0% loss)
Round-trip time	Minimum 0 ms, Maximum 0 ms, Average 0 ms
TTL	128
Bytes per packet	32

2.6 Comparison of the Two Connections

Feature	Point-to-point (Part A)	Star (Part B)
Cable used	Crossover	Straight-through
Central device	None	Switch1 (2960-24TT)
Devices supported	Two	Many, one per switch port
Effect of a cable failure	Link is lost	Only that device is affected

2.7 Result

A point-to-point connection using a crossover cable and a star topology using a switch were implemented in Packet Tracer. In both cases the ping succeeded with 0% packet loss, which verifies connectivity in each topology.